

Chapter 2

Hypothesis Testing

[Hypothesis\(es\) Testing | Springer Nature Link](#)

Learning Objectives

- Understand the importance of stationarity in time series
- Learn the steps in hypothesis testing
- Master the Gauss-Markov assumptions
- Understand diagnostic tests for regression models
- Learn about t-tests and F-tests
- Identify and handle model violations

Stationarity in Time Series

Why is Stationarity Important?

- Many statistical models require stationarity
- Enables effective predictions
- Provides consistent patterns without trends or seasonality

Unit Root

- Feature indicating non-stationarity
- Series has a root of 1 in characteristic equation
- Series wanders randomly without returning to long-term mean
- Difference stationary: Differencing can make it stationary

Stationarity Tests

Augmented Dickey-Fuller (ADF) Test

- Null Hypothesis (H_0): Series has a unit root (non-stationary)
- Alternative (H_1): Series is stationary

KPSS Test

- Null Hypothesis (H_0): Series is stationary
- Alternative (H_1): Series has a unit root (non-stationary)

Steps in Hypothesis Testing

Step 1: Formulate Hypotheses

- Null Hypothesis (H_0): No effect, coefficient = 0 ($\beta = 0$)
- Alternative Hypothesis (H_1): Effect exists, coefficient $\neq 0$ ($\beta \neq 0$)

Step 2: Estimate Model (OLS): Obtain coefficient estimates ($\hat{\beta}$)

Step 3: Calculate Test Statistic: $t = \frac{\hat{\beta} - \beta_0}{SE(\hat{\beta})}$

Step 4: Make Decision

- Compare p-value to significance level (α)
- Reject H_0 if p-value $< \alpha$ (typically 0.05)

Understanding P-Values

Significance Levels (α)

Level	Confidence	Usage
0.10	90%	Less strict
0.05	95%	Standard
0.01	99%	More strict

Decision Rule	Type I Error
• $p\text{-value} < \alpha$: Reject H_0 (statistically significant) • $p\text{-value} \geq \alpha$: Fail to reject H_0	• Rejecting H_0 when it is true • Probability = α

Gauss-Markov Assumptions

Under certain conditions, OLS estimators are BLUE (Best Linear Unbiased Estimators)

The 5 Key Assumptions:

- Linearity: Linear relationship between X and Y
- No Endogeneity: Independent variables not correlated with error term
- Normality & Homoscedasticity: Errors normally distributed with constant variance
- No Autocorrelation: Errors are independent
- No Multicollinearity: Independent variables not highly correlated

Endogeneity

Definition: Independent variable correlated with the error term

Sources of Endogeneity:

- **Omitted Variables**

- Missing variable affects both dependent and independent variables
- Creates spurious relationships

- **Measurement Error**

- Inaccurate measurement obscures true relationship
- Leads to biased estimates

- **Simultaneity**

- Two-way causation between variables
- Dependent variable influences independent variable

Multicollinearity

Definition: Independent variables are highly correlated with each other

Problems Caused:

- Cannot determine individual effects
- Inflated standard errors
- Unreliable coefficient estimates

Detection Methods:

- Variance Inflation Factor (VIF): $VIF > 5$ or 10 indicates high multicollinearity
- Correlation Matrix: Examine correlations among predictors

Diagnostic Tests for Model Validation

Test Category	Purpose	Common Tests
Multicollinearity	Check predictor correlations	VIF, Correlation Matrix
Linearity	Ensure linear relationship	Residuals vs Fitted, Partial Regression
Normality	Check residual distribution	Q-Q Plot, Shapiro-Wilk, Jarque-Bera
Homoscedasticity	Constant variance	Breusch-Pagan, White's Test
Independence	No autocorrelation	Durbin-Watson, Residuals over time
Outliers	Identify influential points	Cook's Distance, Leverage
Model Fit	Overall performance	R^2 , Adjusted R^2 , AIC, BIC

Testing Normality of Residuals

Why Normality Matters

- Stock returns assumed normal
- Error terms should have zero mean
- Important for reliable inference in small samples

Real-World Deviations:

- Heavy Tails: Extreme values occur more frequently
- Skewness: Asymmetric distributions
- Volatility Clustering: Periods of high/low variability
- Leptokurtosis: Sharp peaks with fat tails

Tests:

- Shapiro-Wilk Test
- Jarque-Bera Test
- D'Agostino-Pearson Test

T-Test and F-Test

T-Test

- Tests one coefficient at a time
- Checks if coefficient $\neq 0$
- Based on t-distribution

- **F-Test**

- Tests multiple coefficients simultaneously
- Checks overall model significance
- Based on F-distribution

Bartlett's Test for Equal Variances

Purpose: Checks if variances of several groups are equal

Hypotheses:

- H_0 : Variances are equal across all groups
- H_1 : At least one group has different variance

Autocorrelation Test

Definition: Error terms correlated with each other

Durbin-Watson Test

Value	Interpretation
≈ 2	No autocorrelation
< 2	Positive autocorrelation
> 2	Negative autocorrelation

Ramsey RESET Test

Purpose: Tests if functional form is correctly specified

What RESET Detects:

- Omitted variables
- Incorrect functional forms
- Missing interaction terms
- Nonlinearity

Interpretation:

- $p\text{-value} > 0.05$: No evidence of misspecification

Outlier Detection with Cook's Distance

Purpose: Identify influential observations

Formula:

$$D_i = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(i)})^2}{p \times MSE}$$

Rule of Thumb:

Influential if: $D_i > \frac{4}{n}$ or $D_i > 1$

Parameter Stability Tests

Chow Test

- Detects structural breaks at known point
- Compares subsets before and after break

p-value < 0.05: Significant structural break

CUSUM Test

Purpose: Monitors cumulative sum of residuals for gradual changes

Interpretation:

- $p\text{-value} = 0.139 > 0.05$
- Parameter stability confirmed
- Relationship remains consistent over time

BDS Test

Purpose: Detects serial dependence and nonlinearity

What BDS Tests:

- Nonlinear patterns
- Complex dynamics
- Serial dependence

Interpretation:

- $p\text{-value} = 0.500 > 0.05$
- Linearity confirmed
- No evidence of nonlinearity

ARCH Test for Conditional Heteroscedasticity

Purpose: Detects time-varying variance

Key Concept:

- Variance depends on past error terms
- Volatility clustering in financial data

Interpretation:

- $p\text{-value} = 0.333 > 0.05$
- No significant heteroscedasticity

Bai-Perron Structural Break Test

Purpose: Detects multiple structural breaks

Applications:

- Financial Crises: Changes in stock-market relationships
- Policy Changes: Impact of new regulations
- Industry Shifts: Technological disruptions
- Long Time Series: Multiple economic events

When to Use:

- Analyzing long time series
- Suspect time-dependent relationships
- Multiple potential breakpoints

Diagnostic Tests Summary

Test	Purpose	Good Outcome
Residuals vs Fitted	Linearity	Random scatter
Shapiro-Wilk	Normality	$p > 0.05$
Durbin-Watson	Autocorrelation	≈ 2
Breusch-Pagan	Homoscedasticity	$p > 0.05$
RESET	Specification	$p > 0.05$
Cook's Distance	Outliers	$< 4/n$
Chow Test	Structural Breaks	$p > 0.05$
CUSUM	Parameter Stability	$p > 0.05$
BDS	Nonlinearity	$p > 0.05$
ARCH	Conditional Variance	$p > 0.05$

Questions & Discussion

Review Questions:

- What is stationarity and why is it important?
- What are the five Gauss-Markov assumptions?
- How do you interpret the Durbin-Watson statistic?
- What does Cook's Distance measure?
- When would you use a Chow test?

Discussion Points:

- Impact of violating assumptions on model validity
- Handling influential observations in practice
- When diagnostic tests might be less relevant
- Balancing model complexity and interpretability